Introduction to C#
A Beginner-Friendly Guide to Modern Programming with C#

Author: Luv, Bill

Table of Contents

Introduction

C# (pronounced “C-Sharp”) is a modern, powerful programming language developed by Microsoft. It is widely used for building:

- desktop applications
- web applications (ASP.NET)
- mobile apps (Xamarin / .NET MAUI)
- games (Unity)
- enterprise software

This ebook is designed for beginners who want to learn programming using C# from scratch and build a strong foundation in software development.

Chapter 1 — Introduction to C#

C# is an object-oriented programming language built on the .NET framework.

Why Learn C#?

- Easy to learn for beginners
- Powerful and scalable
- Used in professional environments
- Strong community and support

- Works across platforms with .NET

What You Can Build with C#

- Games using Unity
- Web APIs and websites
- Desktop applications
- Mobile apps
- Cloud applications

Chapter 2 — Setting Up the Development Environment

To start programming in C#, you need:

- .NET SDK
- Visual Studio or Visual Studio Code

Download .NET

.NET Official Website

Recommended IDE

- Visual Studio Community (best for beginners)
- Visual Studio Code (lightweight option)

Chapter 3 — Your First C# Program

```
using System;

class Program
{
   static void Main()
   {
      Console.WriteLine("Hello, World!");
   }
}
```

Explanation

- Main() is the entry point
- Console.WriteLine() prints output

Chapter 4 — Variables and Data Types

Variables store data.

Example

```
int age = 25;
string name = "Chris";
double price = 10.5;
bool isActive = true;
```

Common Data Types

Type	Description
int	Whole numbers
string	Text
double	Decimal numbers
bool	True/false

Chapter 5 — Operators in C#
Arithmetic Operators

```
int sum = 5 + 3;
```

Comparison Operators

```
bool result = 5 > 3;
```

Logical Operators

```
bool result = true && false;
```

Chapter 6 — Control Flow
If Statement

```
int age = 18;
```

```
if (age >= 18)
{
   Console.WriteLine("Adult");
}
```

Switch Statement

```
int day = 1;

switch(day)
{
   case 1:
      Console.WriteLine("Monday");
      break;
}
```

Loops
For Loop

```
for(int i = 0; i < 5; i++)
{
   Console.WriteLine(i);
}
```

Chapter 7 — Methods and Functions

```
static void SayHello()
{
   Console.WriteLine("Hello");
}
```

Method with Parameters

```
static int Add(int a, int b)
{
   return a + b;
}
```

Chapter 8 — Object-Oriented Programming (OOP)

C# is based on OOP principles.

Core Principles

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

Chapter 9 — Classes and Objects

Class Example

```
class Person
{
   public string name;
   public int age;
}
```

Object Creation

```
Person p = new Person();
p.name = "Lena";
```

Chapter 10 — Inheritance and Polymorphism

Inheritance

```
class Animal
{
   public void Eat()
   {
      Console.WriteLine("Eating...");
   }
}

class Dog : Animal
{
}
```

Polymorphism

```
class Animal
{
   public virtual void Sound()
   {
      Console.WriteLine("Animal sound");
   }
}
```

Chapter 11 — Encapsulation and Abstraction

Encapsulation

```
class BankAccount
{
   private double balance;
}
```

Abstraction

```
abstract class Shape
{
   public abstract void Draw();
}
```

Chapter 12 — Arrays and Collections
Array Example

```
int[] numbers = {1, 2, 3};
```

List Example

```
List<string> names = new List<string>();
names.Add("Chris");
```

Chapter 13 — Exception Handling

```
try
{
   int result = 10 / 0;
}
catch(Exception ex)
{
   Console.WriteLine(ex.Message);
}
```

Chapter 14 — File Handling

```
using System.IO;

File.WriteAllText("file.txt", "Hello C#");
```

Chapter 15 — Introduction to LINQ

LINQ helps query data easily.

Example

```
var result = numbers.Where(n => n > 2);
```

Chapter 16 — Building Simple Console Applications

Example: Calculator

```
int a = 5;
int b = 3;
Console.WriteLine(a + b);
```

Chapter 17 — Best Practices in C#

- Use meaningful variable names
- Write clean and readable code
- Avoid duplication
- Follow OOP principles
- Handle exceptions properly

Chapter 18 — Conclusion

C# is a powerful and versatile programming language that opens doors to many areas of software development. By learning the fundamentals covered in this ebook, beginners can build a strong foundation and progress toward advanced development in web, desktop, and game programming.

Final Advice

- Practice coding every day
- Build small projects
- Understand OOP deeply
- Learn .NET ecosystem
- Keep improving step by step

End

Introduction to C# — Beginner Programming Guide